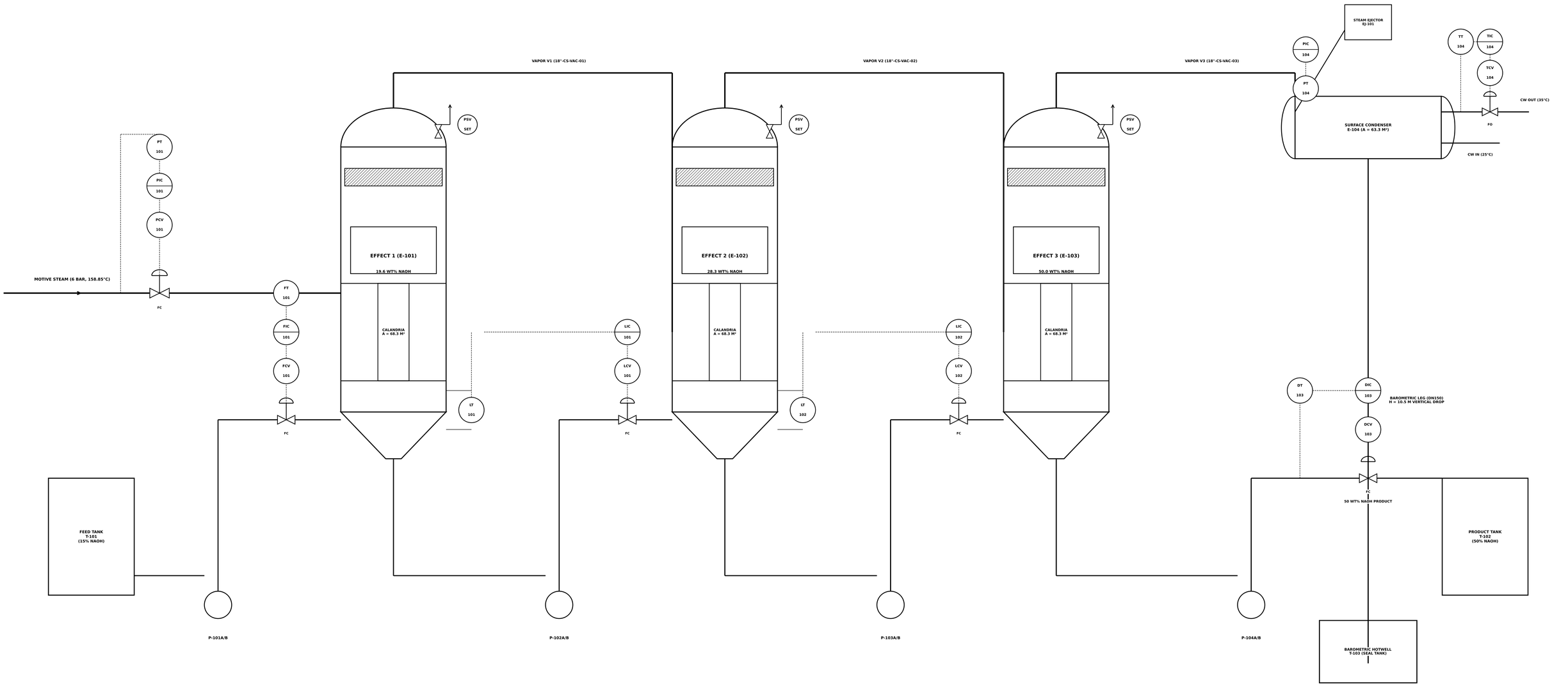
TRIPLE-EFFECT NAOH EVAPORATOR UNIT — DETAILED P&ID

FULL INSTRUMENTATION, DCS CONTROL LOOPS, INTERLOCKS, SAFETY RELIEF VALVES & PIPING CLASSES



OPTIONAL DIAGRAM 2 — DETAILED PROCESS PIPING & INSTRUMENTATION DIAGRAM (P&ID)

DRAWING TITLE PIPING & INSTRUMENTATION DIAGRAM (P&ID)	**SUBJECT** PROCESS HEAT TRANSFER (PHT)	**COURSE / BRANCH** B.E. CHEMICAL ENGINEERING	**INSTRUMENTATION CODE** ISA-5.1 STANDARD SYMBOLS
STUDENT NAME SOURISH SENAPATI	**EXAM ROLL NUMBER** 002310301035	**ASSIGNED ROLL (LAST 2 DIGITS)** 35 (TARGET: 50.0 WT% NAOH PRODUCT)	**DATE** 01 / 09 / 2026
CONTROL PHILOSOPHY AUTOMATIC DCS CASCADE CONTROL	**CONTROL LOOPS** PIC-101, FIC-101, LIC-101/102/103, DIC-103, TIC-104, PIC-104	**FEED CAPACITY / PRODUCT** 400 T/D FEED (15% NAOH) → 5,000 KG/H PRODUCT (50% NAOH)	**PLANT DESIGNATION** TRIPLE-EFFECT FORWARD-FEED CAUSTIC EVAPORATION UNIT

STANDARDS: ANSI/ISA-5.1-2009, ASME B31.3 PROCESS PIPING | ALL SIGNALS: 4-20 MA / SMART HART / FIELDBUS